Tenta HT 23 (1)

SF2940

1) a.) Let $A, B \in \mathcal{A}$. Then $A \cup B = (A \cap B) \cup (A \cap B^c) \cup (A^c \cap B)$.

We observe that the events $(A \cap B)$, $(A \cap B^c)$ and $(A^c \cap B)$ are disjoint. By additivity of probability measures,

$$\mathbb{P}(A \cup B) = \underbrace{\mathbb{P}(A \cap B) + \mathbb{P}(A \cap B^c)}_{= \mathbb{P}(A) \text{ since } (A \cap B) \text{ and } (A \cap B^c) \text{ are disjoint}} + \mathbb{P}(A^c \cap B).$$

Now we note that $(A^c \cap B) \cup (A \cap B) = B$, since the events on the left side are disjoint, we have by additivity $\mathbb{P}(A^c \cap B) + \mathbb{P}(A \cap B) = \mathbb{P}(B)$.

$$\Rightarrow \mathbb{P}(A^c \cap B) = \mathbb{P}(B) - \mathbb{P}(A \cap B).$$

It follows that $\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B) - \mathbb{P}(A \cap B)$. //

b) Let $\bar{Y}$ be another random variable that has the same distribution as $\bar{X}$, but is independent from $\bar{X}$. Then

$$\varphi_{\bar{X}+\bar{Y}}(t) \overset{\text{indep}}{=} \varphi_{\bar{X}}(t)\,\varphi_{\bar{Y}}(t) = (\varphi_{\bar{X}}(t))^2$$

since $\bar{X}$ and $\bar{Y}$ have the same characteristic function.

c.) For ease of notation let $Y_i := \mathbb{1}\{X_i \le x\}$. Since X_i are independent, so are Y_i.

The expected value of each Y_i is $\mathbb{E}[Y_i] = \mathbb{E}[\mathbb{1}\{X_i \le x\}] = \mathbb{P}(X_i \le x) = F_{X_i}(x) =: F(x)$.

Since X_i are identically distributed, we have $\mathbb{E}\hat{F}_n(x) = \frac{1}{n}\sum_{i=1}^{n} \mathbb{E}[Y_i] = F_{X_1}(x) = F(x)$.

Consider next,

$$\mathbb{E}\left[(\hat{F}_n(x) - F(x))^2\right] = \mathbb{E}\left[\left(\frac{1}{n}\sum_{i=1}^{n}(Y_i - \mathbb{E}[Y_i])\right)^2\right]$$

$$= \mathbb{E}\left[\frac{1}{n^2}\sum_{i=1}^{n}\sum_{k=1}^{n}(Y_i - \mathbb{E}[Y_i])(Y_k - \mathbb{E}[Y_k])\right]$$

$$= \frac{1}{n^2} \sum_{i=1}^{n} \mathbb{E}\left[(Y_i - \mathbb{E}[Y_i])^2 \right] + \frac{1}{n^2} \sum_{i \neq k} \mathbb{E}\left[(Y_i - \mathbb{E}[Y_i])(Y_k - \mathbb{E}[Y_k]) \right]$$

independence: $= \mathbb{E}[Y_i - \mathbb{E}[Y_i]] \cdot \mathbb{E}[Y_k - \mathbb{E}[Y_k]]$, where $\mathbb{E}[Y_i - \mathbb{E}[Y_i]] = 0$ and $\mathbb{E}[Y_k - \mathbb{E}[Y_k]] = 0$

$$= \frac{1}{n^2} \sum_{i=1}^{n} \mathrm{Var}(Y_i)$$

identically distributed

$$= \frac{1}{n} \mathrm{Var}(Y_1).$$

Yet the variance of Y_1 is bounded by one, as Y_1 takes values zero and one. In detail,

$$\mathrm{Var}\, Y \leq \mathbb{E}[Y_1^2] = 0 \cdot \mathbb{P}(Y_1 = 0) + 1^2 \mathbb{P}(Y_1 = 1) \leq 1.$$

Thus $$\mathbb{E}\left[\left(\hat{F}_n(x) - F(x) \right)^2 \right] \leq \frac{1}{n},$$

and we have $\hat{F}_n(x) \xrightarrow[n \to \infty]{2} F(x)$, where $F(x)$ is the common cumulative distribution function the (X_i)'s. //

Short cut: (Y_i) are i.i.d. with $\mathrm{Var}\, Y_i \leq 1$. Indeed, $\mathrm{Var}\, Y_i = \mathbb{E}[Y_i^2] - \mathbb{E}[Y_i]^2$

Y_i takes values 0 and 1, hence $Y_i^2 = Y_i$.

$$= \mathbb{E}[Y_i] - \mathbb{E}[Y_i]^2$$

$$= F(x) - F(x)^2$$

$$= F(x)(1 - F(x)) \leq 1,$$

since $0 \leq F(x) \leq 1$.

By the weak law of large numbers

$$\frac{1}{n} \sum_{i=1}^{n} Y_i \xrightarrow[n \to \infty]{2} \mathbb{E}[Y_1] = F_{X_1}(x) = F(x).$$

//

2 a.) Let $B = \begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}$. Then

$$\begin{pmatrix} Y_1 \\ Y_2 \end{pmatrix} = \begin{pmatrix} X_1 + X_3 \\ X_2 \end{pmatrix} = B \begin{pmatrix} X_1 \\ X_2 \\ X_3 \end{pmatrix}.$$

Since all X_i have zero expectation, the Y_i also have zero expectation. The covariance matrix $\Lambda_{\vec{Y}}$ is, according to the linear transformation theorem given by $\Lambda_{\vec{Y}} = B \Lambda_{\vec{X}} B^T$. Thus

$$\Lambda_{\vec{Y}} = \begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix} \begin{pmatrix} 1 & 1 & 1 \\ 1 & 2 & 2 \\ 1 & 2 & 3 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 0 \end{pmatrix} = \begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix} \begin{pmatrix} 2 & 1 \\ 3 & 2 \\ 4 & 2 \end{pmatrix}$$

$$= \begin{pmatrix} 6 & 3 \\ 3 & 2 \end{pmatrix}.$$

Hence $\vec{Y} \in \mathcal{N}\left(0, \begin{pmatrix} 6 & 3 \\ 3 & 2 \end{pmatrix}\right)$.

Our goal is to compute

$$f_{Y_2 | Y_1 = y_1}(y_2) = \frac{f_{(\vec{Y}_1, \vec{Y}_2)}(y_1, y_2)}{f_{\vec{Y}_1}(y_1)}.$$

Hence, we need to invert $\Lambda_{\vec{Y}}$, by Cramer's rule

$$\Lambda_{\vec{Y}}^{-1} = \frac{1}{3} \begin{pmatrix} 2 & -3 \\ -3 & 6 \end{pmatrix} = \begin{pmatrix} \frac{2}{3} & -1 \\ -1 & 2 \end{pmatrix}.$$

Thus $$f_{(Y_1,Y_2)}(y_1,y_2) = \frac{1}{2\pi\sqrt{3}} e^{-\frac{1}{2}(y_1,y_2)\begin{pmatrix} \frac{2}{3} & -1 \\ -1 & 2 \end{pmatrix}\begin{pmatrix} y_1 \\ y_2 \end{pmatrix}}$$

$$= \frac{1}{2\pi\sqrt{3}} e^{-\frac{1}{2}\left(\frac{2}{3}y_1^2 - 2y_1\cdot y_2 + 2y_2^2\right)}$$

From the above, we know that $Y_1 \in N(0,6)$, hence $f_{y_1}(y_1) = \frac{1}{\sqrt{2\pi}\sqrt{6}} e^{-\frac{1}{12}y_1^2}$

Therefore,

$$f_{y_2|y_1=y_1}(y_2) = \frac{\frac{1}{2\pi\sqrt{3}} e^{-\frac{1}{3}y_1^2 + \frac{1}{12}y_1^2 + y_1\cdot y_2 - y_2^2}}{\frac{1}{\sqrt{2\pi}\sqrt{6}}}$$

$$= \frac{\sqrt{2}}{\sqrt{2\pi}} e^{-\frac{1}{4}y_1^2 + y_1\cdot y_2 - y_2^2}$$

$$= \frac{1}{\sqrt{\pi}} e^{-(y_2 - \frac{1}{2}y_1)^2} \quad //$$

2c.) We introduce the variables $\bar{U} = \frac{\bar{X}_1}{\bar{X}_2}$, $\bar{V} = \bar{X}_2$.

Then $\bar{X}_1 = \bar{U}\bar{V}$, $\bar{X}_2 = V$.

The Jacobian of $(u,v) \to (uv, v)$ is

$$J = \begin{pmatrix} v & u \\ 0 & 1 \end{pmatrix} = v,$$

thus the joint density of $(\bar{U}, \bar{V})$ is

$$f_{\bar{U},\bar{V}}^{(u,v)} = f_{\bar{X}_1,\bar{X}_2}(uv, v)\,|v|, \quad u,v \in \mathbb{R}.$$

The covariance matrix of $(X_1, X_2)^T$ is $\Lambda = \begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix}$.

Its inverse is $\Lambda^{-1} = \begin{pmatrix} 2 & -1 \\ -1 & 1 \end{pmatrix}$, and $\det \Lambda = 1$.

Thus,

$$f_{\bar{X}_1,\bar{X}_2}(x_1,x_2) = \frac{1}{2\pi} e^{-\frac{1}{2}(x_1,x_2)\begin{pmatrix} 2 & -1 \\ -1 & 1 \end{pmatrix}\begin{pmatrix} x_1 \\ x_2 \end{pmatrix}}$$

$$= \frac{1}{2\pi} e^{-\frac{1}{2}\left(2x_1^2 - 2x_1 x_2 + x_2^2\right)}.$$

Setting $x_1 = uv$ and $x_2 = v$, we get

$$f_{\bar{X}_1,\bar{X}_2}(uv, v) = \frac{1}{2\pi} e^{-\frac{1}{2}\left(2u^2v^2 - 2uv^2 + v^2\right)}$$

$$= \frac{1}{2\pi} e^{-\frac{v^2}{2}(2u^2 - 2u + 1)}.$$

Thus $$f_{\bar{U},\bar{V}}(u,v) = \frac{1}{2\pi} e^{-\frac{v^2}{2}(2u^2-2u+1)} |v|.$$

Since $\bar{U} = \frac{\bar{X}_1}{\bar{X}_2}$ we get the distribution of $\frac{\bar{X}_1}{\bar{X}_2}$ by integrating out v:

$$f_{\bar{U}}(u) = \frac{1}{2\pi}\int_{\mathbb{R}} |v| e^{-\frac{v^2}{2}(2u^2-2u+1)} dv$$

$$= \frac{1}{\pi}\int_0^\infty v\, e^{-\frac{v^2}{2}(2u^2-2u+1)}$$

$$= \frac{1}{\pi}\int_0^\infty v\, e^{-\frac{v^2}{2}a}, \text{ with } a := (2u^2-2u+1).$$

Note that $$v e^{-\frac{v^2}{2}a} = -\frac{1}{a}\frac{\partial}{\partial v} e^{-\frac{v^2}{2}a}, \text{ hence}$$

$$\frac{1}{\pi}\int_0^\infty v e^{-\frac{v^2}{2}a} dv = -\frac{1}{\pi a}\int_0^\infty \frac{\partial}{\partial v}\left(e^{-\frac{v^2}{2}a}\right) dv$$

$$= -\frac{1}{\pi a}\left(e^{-\frac{v^2}{2}a}\right)\Big|_0^\infty.$$

Clearly, $a = u^2 - 2u + 1 + u^2 = (u-1)^2 + u^2 > 0$, $u \in \mathbb{R}$.

Thus $$\frac{1}{\pi}\int_0^\infty v e^{-\frac{v^2}{2}a} dv = \frac{1}{\pi a}.$$

It follows that

$$f_{\bar{U}}(u) = \frac{1}{\pi} \frac{1}{(u-1)^2 + u^2}, \quad u \in \mathbb{R}.$$

Absurd:

$$f_{\frac{\bar{X}_1}{\bar{X}_2}}(u) = \frac{1}{\pi} \frac{1}{(u-1)^2 + u^2} = \frac{1}{\pi} \frac{\frac{1}{2}}{(u-\frac{1}{2})^2 + (\frac{1}{2})^2}, \quad u \in \mathbb{R},$$

which is the density of the Cauchy law $C(\frac{1}{2}, \frac{1}{2})$. //

3.) a.) The probability generating function of X_1 (and for all other X_i's) is

$$g_{X_1}(t) = \sum_{k=1}^{\infty} t^k p_{X_1}(k) = \sum_{k=1}^{\infty} t^k \frac{m^k}{k!} \frac{1}{e^m - 1}$$

$$= \frac{1}{e^m - 1} \sum_{k=1}^{\infty} \frac{(tm)^k}{k!} = \frac{1}{e^m - 1}\left(\sum_{k=0}^{\infty} \frac{(tm)^k}{k!} - 1 \right)$$

$$= \frac{1}{e^m - 1}\left(e^{tm} - 1 \right)$$

$$= \frac{e^{-m}}{1 - e^{-m}}\left(e^{tm} - 1 \right), \text{ for all } t \in \mathbb{R}.$$

b.) Since N is independent from the (X_i)'s, we can use the composition formula for the probability generating function

$$g_{S_N}(t) = g_N\left(g_{X_1}(t) \right).$$

Since, $p_N(k) = \binom{n}{k} p^k q^{n-k}$, $k = 0, 1, \ldots, n$,

with $p = 1 - e^{-m}$, $q = e^{-m}$, we have

$$g_N(t) = \sum_{k=0}^{n} \binom{n}{k} (tp)^k q^{n-k} = (q + tp)^n$$

$$= \left(e^{-m} + t(1 - e^{-m}) \right)^n.$$

Thus $$g_{S_N}(t) = \left(e^{-m} + g_{X_1}(t)(1-e^{-m}) \right)^n$$

$$= \left(e^{-m} + \frac{e^{-m}}{1-e^{-m}} \left(e^{tm}-1\right)\left(1-e^{-m}\right) \right)^n$$

$$= \left(e^{-m} + e^{tm-m} - e^{-m} \right)^n$$

$$= \left(e^{tm-m} \right)^n$$

$$= e^{n \cdot m (t-1)}, \quad \text{for all } t \in \mathbb{R}.$$

This is the probability generating function of a Poisson random variable with parameter $n \cdot m$, i.e. $S_N \in Po(n \cdot m)$ by the uniqueness theorem. //

4.) a) Since $\bar{Y}|\bar{X}=x \in Po(x)$, its characteristic function is

$$\varphi_{\bar{Y}|\bar{X}=x}(t) = e^{x(e^{it}-1)}.$$

By the law of total expectation, we have

$$\varphi_{\bar{Y}}(t) = \mathbb{E}[e^{it\bar{Y}}] = \mathbb{E}[\mathbb{E}[e^{it\bar{Y}}|\bar{X}]]$$

$$= \mathbb{E}[e^{\bar{X}(e^{it}-1)}].$$

Since $X \in \Gamma(s,1)$, we have with the hint

$$\varphi_{\bar{Y}}(t) = \mathbb{E}[e^{(e^{it}-1)\bar{X}}] = \frac{1}{(1-(e^{it}-1))^s} = \frac{1}{(2-e^{it})^s}.$$

b.) We first compute $\mathbb{E}[Y]$ and $Var(Y)$. Note that

$$\frac{d}{dt}\varphi_Y(t) = \frac{-s}{(2-e^{it})^{s+1}} \cdot (-i)e^{it} = \frac{is}{(2-e^{it})^{s+1}} e^{it}.$$

For the second derivative, we get

$$\frac{d^2}{dt^2}\varphi_Y(t) = \frac{+is(s+1)}{(2-e^{it})^{s+2}} i e^{it} e^{it} - \frac{s}{(2-e^{it})^{s+1}} e^{it}.$$

Choosing $t=0$, we have

$$\varphi_Y'(0) = is, \quad \varphi_Y''(0) = -s(s+1) - s = -s^2 - 2s.$$

Thus

$$\mathbb{E}[Y] = s, \quad \mathbb{E}[Y^2] = s^2 + 2s, \quad \mathrm{Var}\, Y = 2s.$$

Set now $Z := \dfrac{Y - \mathbb{E}[Y]}{\sqrt{\mathrm{Var}\, Y}} = \dfrac{Y - s}{\sqrt{2s}}$.

Then its characteristic function is

$$\varphi_Z(t) = \mathbb{E}\left[e^{\frac{it}{\sqrt{2s}}(Y-s)}\right] = e^{-i\frac{ts}{\sqrt{2s}}}\,\mathbb{E}\left[e^{\frac{it}{\sqrt{2s}}Y}\right]$$

$$= e^{-it\sqrt{\frac{s}{2}}}\,\varphi_Y\left(\frac{t}{\sqrt{2s}}\right).$$

By part a), we have thus

$$\varphi_Z(t) = e^{-it\sqrt{\frac{s}{2}}}\,\frac{1}{\left(2 - e^{i\frac{t}{\sqrt{2s}}}\right)^s} = e^{-it\sqrt{\frac{s}{2}}}\,\frac{1}{\left(1 - \left(e^{i\frac{t}{\sqrt{2s}}} - 1\right)\right)^s}.$$

Now we want to take $s \to \infty$. Consider therefore

$$\log\left(\frac{1}{\left(1 - \left(e^{i\frac{t}{\sqrt{2s}}} - 1\right)\right)^s}\right) = -s \log\left(1 - \left(e^{i\frac{t}{\sqrt{2s}}} - 1\right)\right).$$

As s becomes large, we have $e^{i\frac{t}{\sqrt{2s}}} - 1 = \dfrac{it}{\sqrt{2s}} - \dfrac{t^2}{4s} + o\left(\dfrac{1}{s}\right)$.

Thus

$$-s \log\left(1 - \left(e^{i\frac{t}{\sqrt{2s}}} - 1\right)\right) = -s \log\left(1 - \frac{it}{\sqrt{2s}} + \frac{t^2}{4s} + o\left(\frac{1}{s}\right)\right)$$

Hint with $z = \dfrac{-it}{\sqrt{2s}} + \dfrac{t^2}{4s} + o\left(\dfrac{1}{s}\right)$

$$= -s\left(\frac{-it}{\sqrt{2s}} + \frac{t^2}{4s} - \frac{1}{2}\left(\frac{it}{\sqrt{2s}}\right)^2 + o\left(\frac{1}{s}\right)\right)$$

$$= -s\left(\frac{-it}{\sqrt{2s}} + \frac{t^2}{2s} + o\left(\frac{1}{s}\right)\right),$$

$$= it\sqrt{\frac{s}{2}} - \frac{t^2}{2} + o(1) \quad , \text{ as } s \to \infty$$

Returning to the characteristic function, we get

$$\varphi_Z(t) = e^{-it\sqrt{\frac{s}{2}}} \cdot e^{it\sqrt{\frac{s}{2}} - \frac{t^2}{2} + o(1)}$$

$$= e^{-\frac{t^2}{2} + o(1)} \quad , \text{ as } s \to \infty.$$

Since the exponential function is continuous, we have

$$\lim_{s \to \infty} \varphi_Z(t) = e^{-\frac{1}{2}t^2}.$$

By the uniqueness theorem, we conclude that $\dfrac{Y - E[Y]}{\sqrt{Var(Y)}} \xrightarrow[s \to \infty]{d} N(0,1)$. //